HDOC-47

Ceil Index in a Sorted Integer Array using Pointers

Problem Statement

Write a C program that reads the size and elements of a sorted variable-length integer array in ascending order, and an integer target. Pass the array to a function named ceil_index() using its address/pointers.

- Find the index of the smallest element in arr[] that is greater than or equal to target.
- Treat that element as the ceil of target.
- If the ceil value occurs multiple times, return the index of its first occurrence.
- Display both the ceil element and its index.
- Display -1 if no element is greater than or equal to target.

Algorithm

1. Read n, then read n sorted integer elements into arr[].
2. Read target.
3. Call ceil_index(arr, n, target). The array is passed as a pointer.
4. Inside ceil_index(), scan from index 0 to n - 1. The first element satisfying *(arr + i) >= target is the ceil; return i immediately.
5. If no such element is found, return -1.
6. In main(), print -1 when the returned index is -1; otherwise print the ceil value and its index.

C Program

```
#include <stdio.h>

int ceil_index(int *arr, int n, int target)
{
   int i;
   for (i = 0; i < n; i++)
   {
      if (*(arr + i) >= target)
         return i;
   }
   return -1;
}

int main()
{
   int n, i, target, index;

   printf("Enter size of array: ");
   scanf("%d", &n);

   int arr[n];
   printf("Enter %d elements in ascending order: ", n);
   for (i = 0; i < n; i++)
      scanf("%d", arr + i);
```

```
    printf("Enter target: ");
    scanf("%d", &target);

    index = ceil_index(arr, n, target);

    if (index == -1)
        printf("-1\n");
    else
        printf("Ceil of %d: %d at index %d\n",
            target, *(arr + index), index);

    return 0;
}
```

Test Case Table

Test Case	Array	Target	Expected Output
1	5 7 7 8 8 10	6	Ceil of 6: 7 at index 1
2	5 7 7 8 9 10	8	Ceil of 8: 8 at index 3
3	5 7 7 8 8 10	11	-1
4	2 4 6 8 10	2	Ceil of 2: 2 at index 0
5	1 3 5 7 9	4	Ceil of 4: 5 at index 2

Result

The program successfully finds the first index of the smallest array element greater than or equal to the target using pointer-based array access. If no ceil element exists, it returns and displays -1.